Roadmap. The first meeting: introductions, the books, logistics. The course follows Srednicki's three parts in his order: spin 0 (LecNote1–15: why relativity needs fields, Lorentz symmetry, canonical quantization, LSZ, path integrals, Feynman rules, one-loop renormalization and the RG), spin $\frac{1}{2}$ (LecNote16–20), and spin 1 with QED (LecNote21–23). Next: LecNote1, Srednicki §1. Supplements: QFTHistory, GroupTheoryIntro.

1 Fall 2025 only. This term's TA, office hours and rooms, and the homework, final and grading policy: see Canvas.

11 9–10
Ishman
Mahbub ①
<mahbub008@
285-03

375-22
W 12:30-13:30

Class 1.

Before entering to the Course material.

Let students introduce themselves. (name, [redacted] major, and why studying QFT).

2 These notes follow Srednicki: $g = \mathrm{diag}(-1,+1,+1,+1)$ (LecNote1 p. 3), so $p^2 = -m^2$ on shell and KG is $(-\partial^2 + m^2)\varphi = 0$. Peskin–Schroeder, Zee and Tong use $(+,-,-,-)$: $p^2 = +m^2$, and $\frac{1}{i}(k^2 + m^2 - i\epsilon)^{-1}$ for $\langle 0|\mathcal{T}\varphi\varphi|0\rangle$ here is their $i(k^2 - m^2 + i\epsilon)^{-1}$. Weinberg: as Srednicki.

Text book:
☆ Mark Srednicki:
other references: Peskin & Schröeder:
A. Zee.
S. Weinberg.

Warning ② matrix representations notations, etc.

Discuss w. friends: M. Schwartz.
D. Tong. (comment on each book & recommend different ones for different levels of students).

HW & Final & Grading: ≠

Confusions & unhappiness are expected.
part of natural learning!

I will follow the textbook. from previous teaching experiences.

Questions about this class?